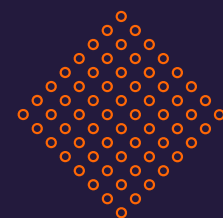


Ecommerce Data Analysis

Logistics Issues in Ecommerce
Delivery to improve profitability

**15% Gain in Delivery
Efficiency**

Addressing Regional Disparities:
Reducing São Paulo's Pending Sales,
Which Are 68% Higher Than Other
States.



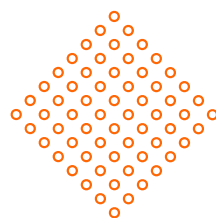
Ecommerce Analysis

By : Daddy Ananta



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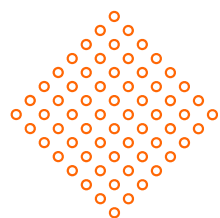
Ecommerce Analysis

Project Goals

The goal of this project is **to identify and address logistics issues in ecommerce delivery that impact profitability.**

*By analyzing delivery inefficiencies, such as delays, shipping challenges, and demand fluctuations across different regions, the project aims **to provide data-driven recommendations for optimizing delivery processes.***

These improvements will enhance customer satisfaction and ultimately increase profitability through better logistics management and increased sales in various regions..



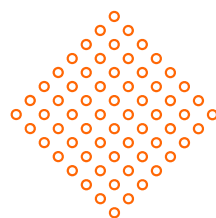
Ecommerce Analysis



Business Understanding

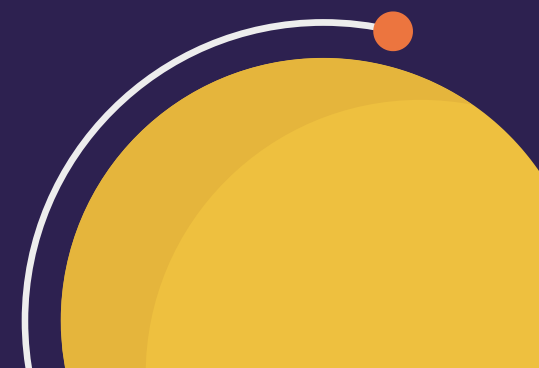
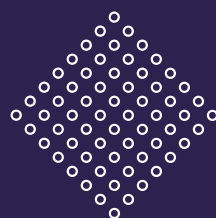
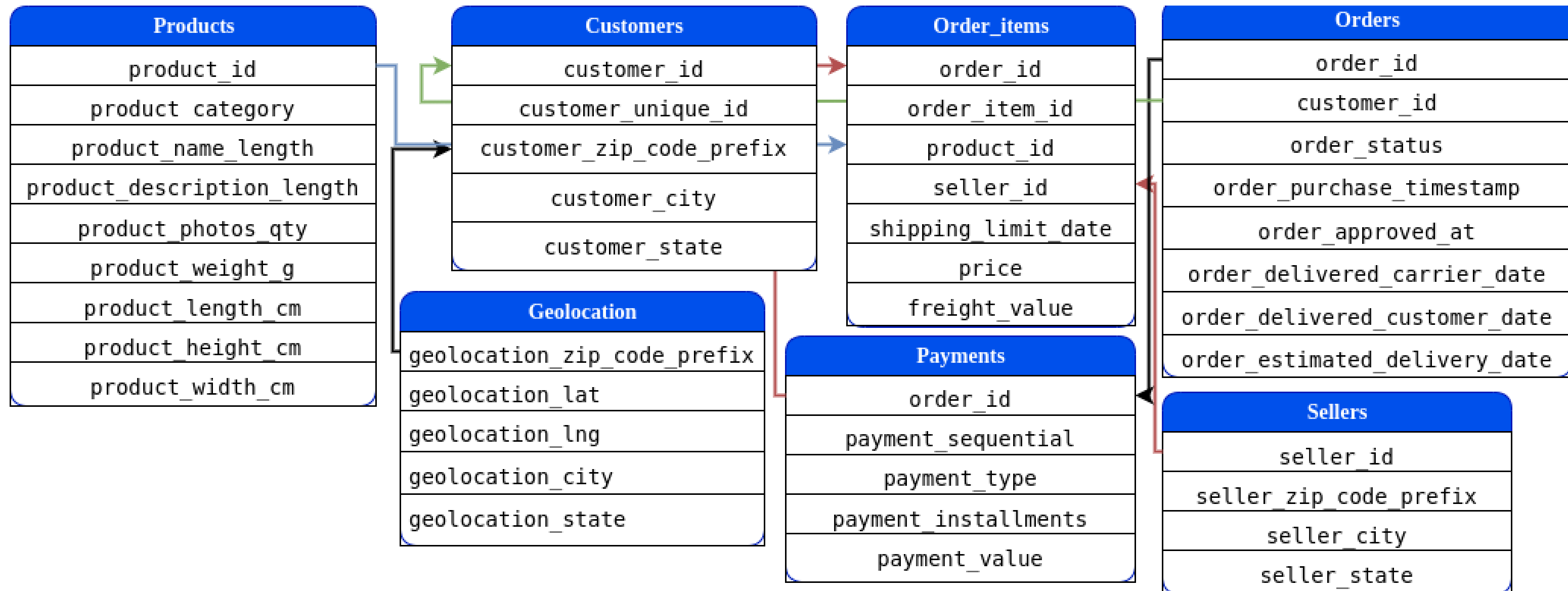
A-Ecommerce (fictional) is a **globally recognized brand and a leading retailer in the United States**, known for offering exceptional value, inspiration, innovation, and a unique shopping experience.

This dataset focuses on A-Ecommerce operations in Brazil, **covering 100,000 orders placed between 2017 and 2018**. It *includes detailed information on order status, pricing, payment and shipping performance, customer locations, product attributes, and customer reviews.*



Ecommerce Analysis

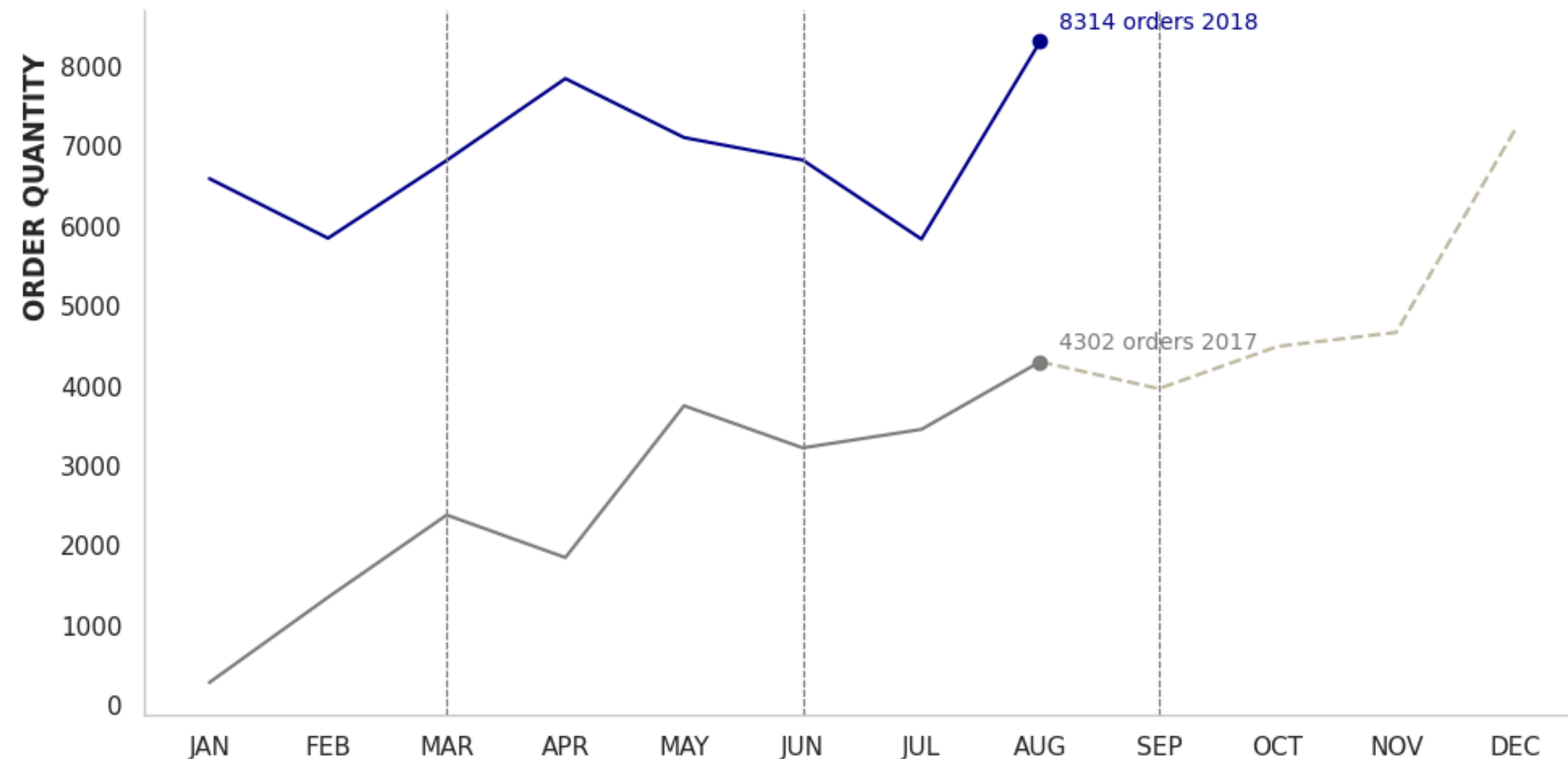
About Dataset



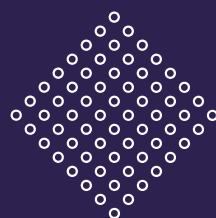
Descriptive Analysis

Orders Purchased **Increased** from 2017 to **2018**

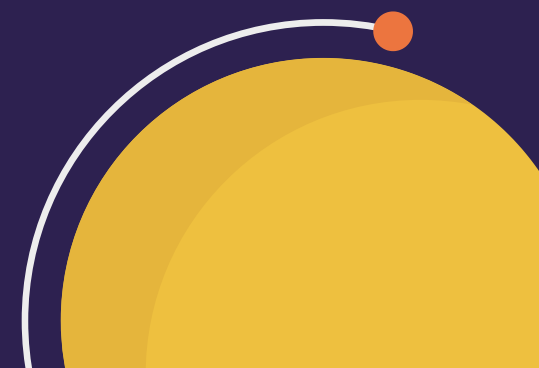
Number of Orders per Month
2017 - 2018



Based on the graph, there is a noticeable increase in purchase demand in 2018 compared to 2017. This increase is influenced by the success of the marketing strategy driven by word of mouth.

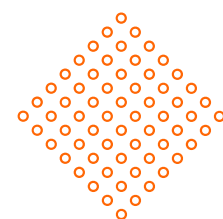
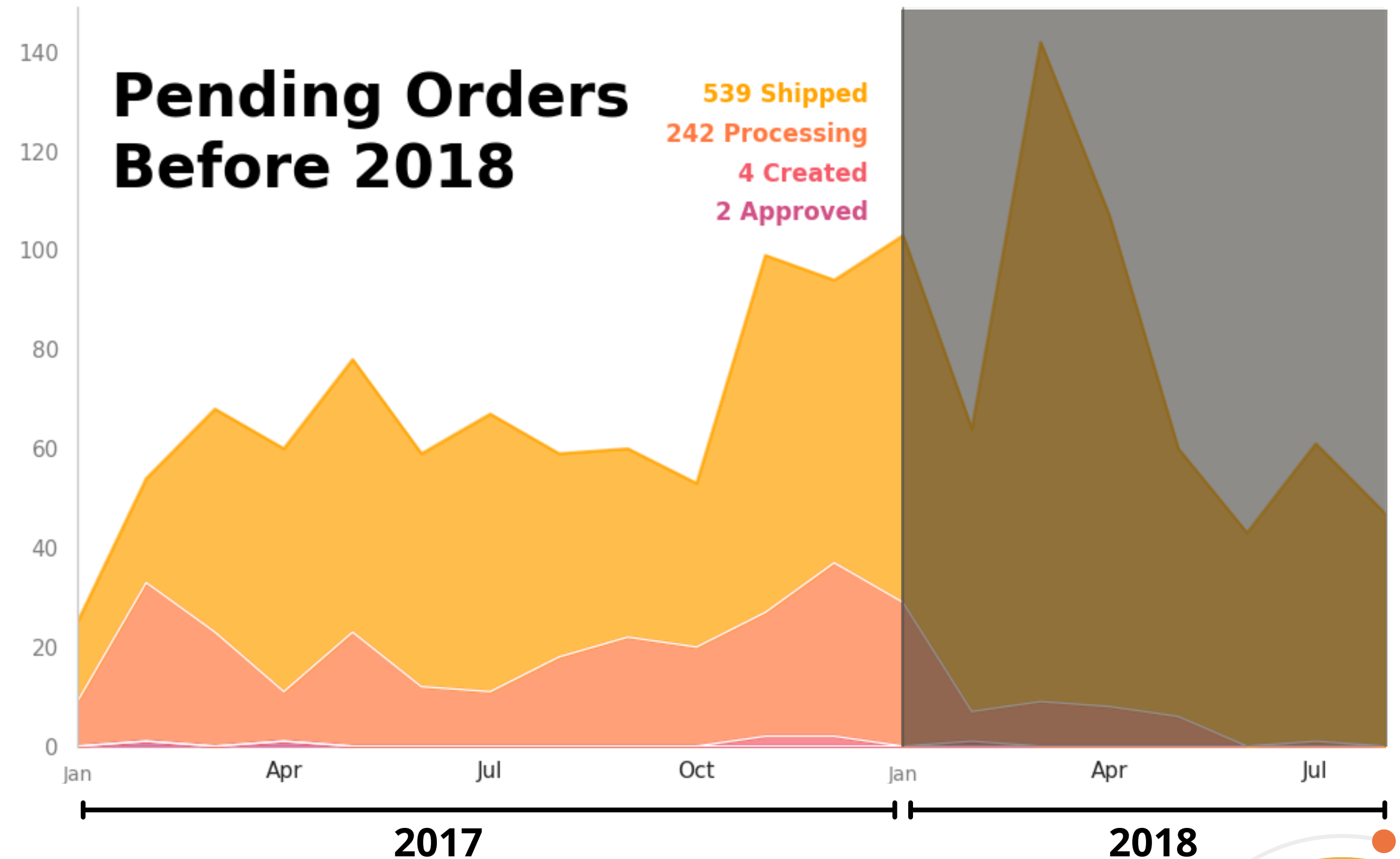


Ecommerce Analysis



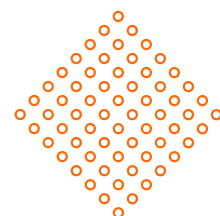
Logistics Challenge

It is evident that before 2018, there were **still orders experiencing delays and not confirmed as delivered**. This issue can lead to customer dissatisfaction, which in turn may affect customer loyalty and give an advantage to competitors.



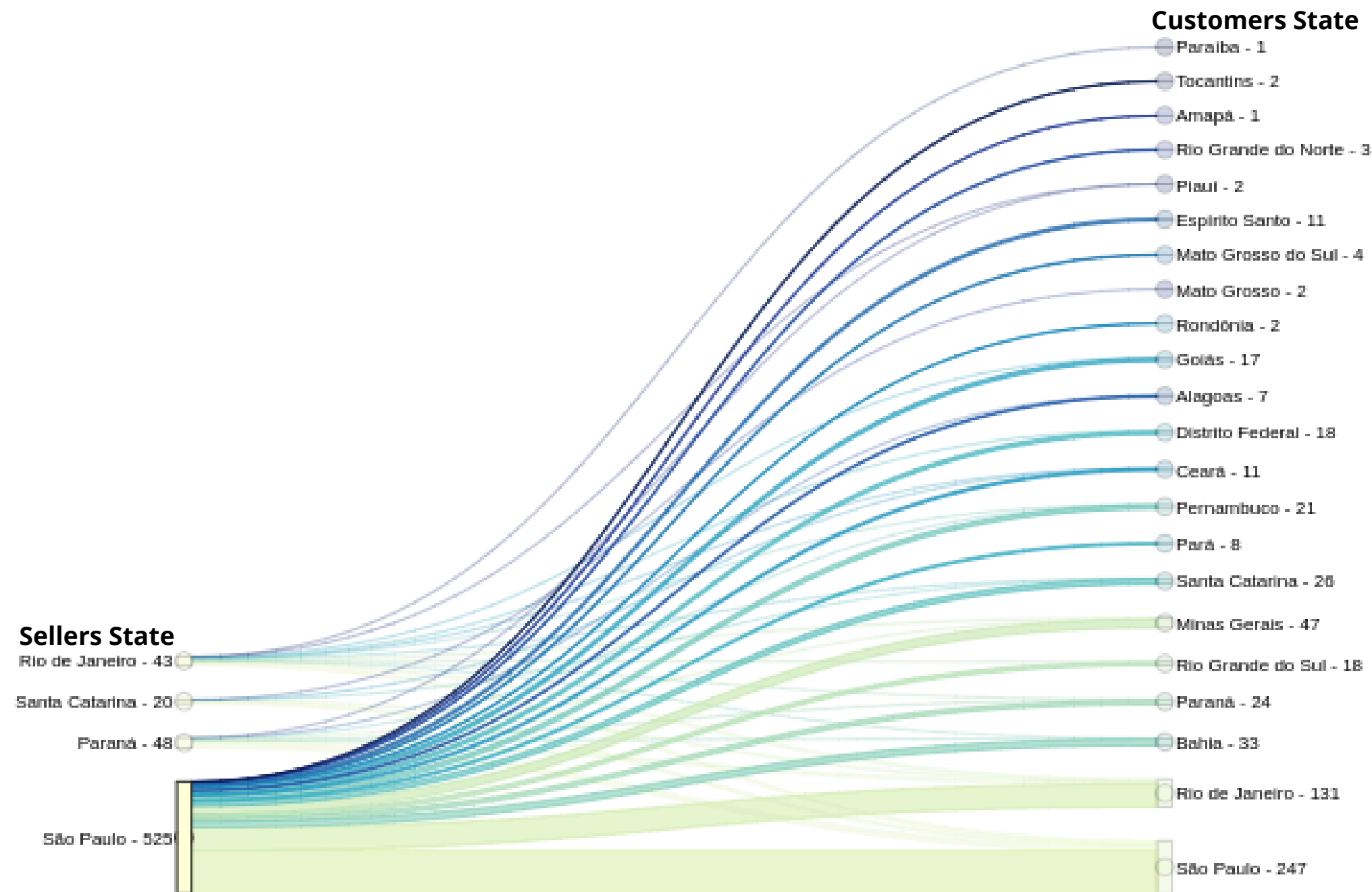
Purchasing Process in E-commerce

The **post-purchase process** in e-commerce involves the steps from order placement to product delivery, including payment approval and shipment. *Each stage of this process impacts the customer experience and influences their likelihood of returning.*

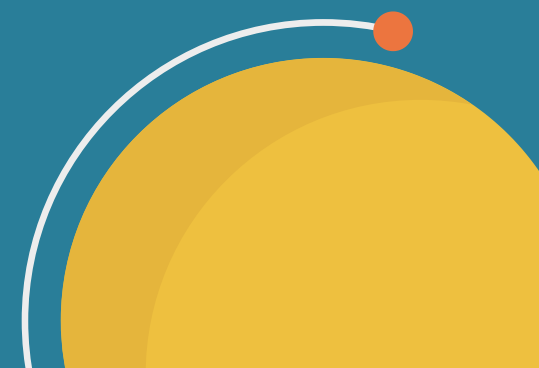
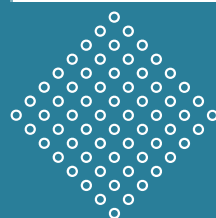


Logistics Challenge

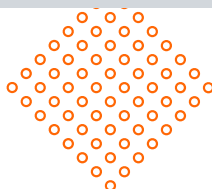
Pending Shipping and Delivery Flow Between
Top 4 Sellers and Consumers in 2017



The number of sales frequently pending is higher in São Paulo (525) compared to Rio de Janeiro (43), Santa Catarina (20), and Paraná (48). **Overall, the percentage of pending sales in São Paulo accounts for 68% of all pending transactions.** What is the main cause of this issue in São Paulo?

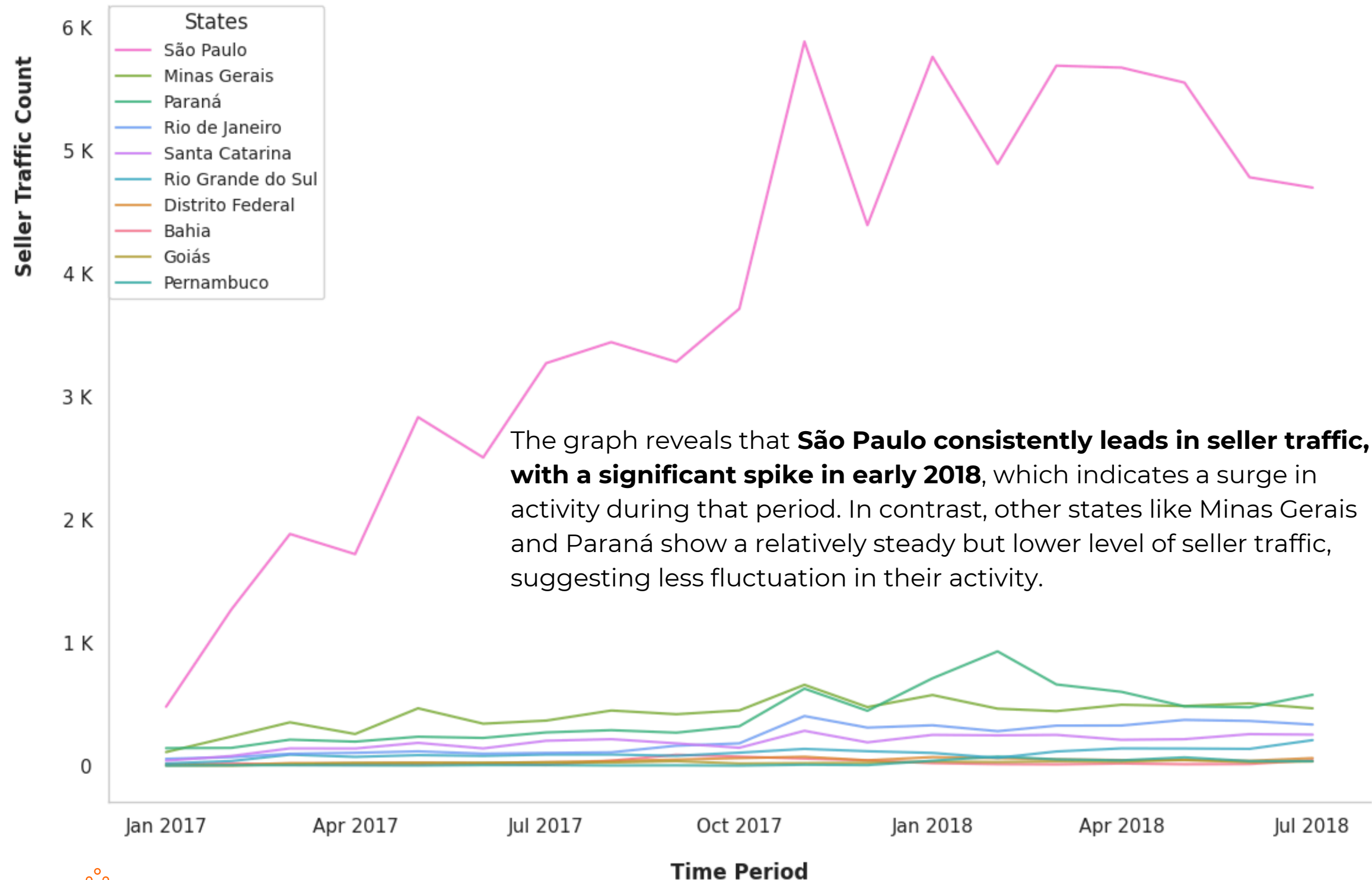


Logistics Challenge



Logistics Challenge

Trend of Seller Traffic in Top 10 States



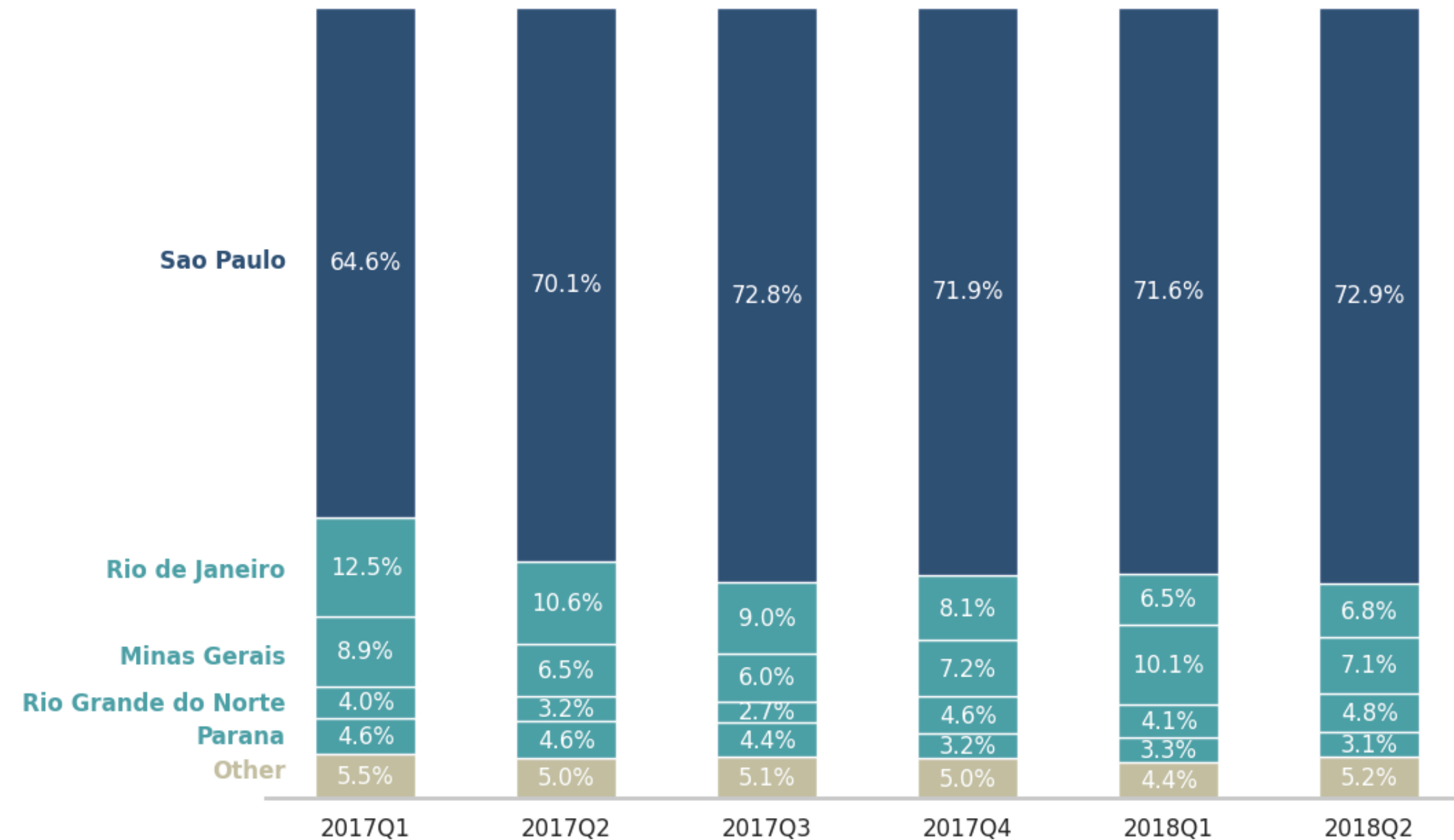
This trend can be likened to a giant monster in the vast ocean, where São Paulo represents the immense power and dominance, rising above the waves, while other states resemble steadier creatures moving along calmer waters, requiring development to grow bigger.

However, the growing presence of São Paulo may put increasing pressure on these smaller states, potentially hindering their growth and even threatening their survival if they are unable to adapt and compete with the surge in activity.

Logistics Challenge

São Paulo's Dominance Grows, While Other States' Shares Shrink

Distribution of Sellers by State in Top 5 and Other Groups



São Paulo's share of sellers consistently increases, reaching 72.9% by Q2 2018, **while other states, such as Rio de Janeiro and Minas Gerais, experience a steady decline in their shares.** This growing dominance of São Paulo suggests an expanding concentration of sellers in the state, **potentially limiting opportunities for growth in other regions.**

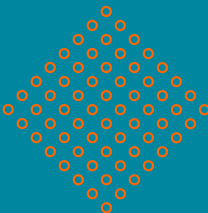
Business Recommendations

In addition to São Paulo, the five other states—**Minas Gerais, Paraná, Rio Grande do Sul, and Santa Catarina**—can serve as solutions to address the heavy reliance on São Paulo. With a significant amount of purchases in these states, they can become alternatives for *redirecting purchases or expanding sales coverage, helping to distribute the transaction load and reduce pressure on São Paulo, while capitalizing on the market potential in these regions.*

São Paulo's Central Role in Seller-Customer Transactions

Heatmap of Orders Between States in January 2017 to August 2018

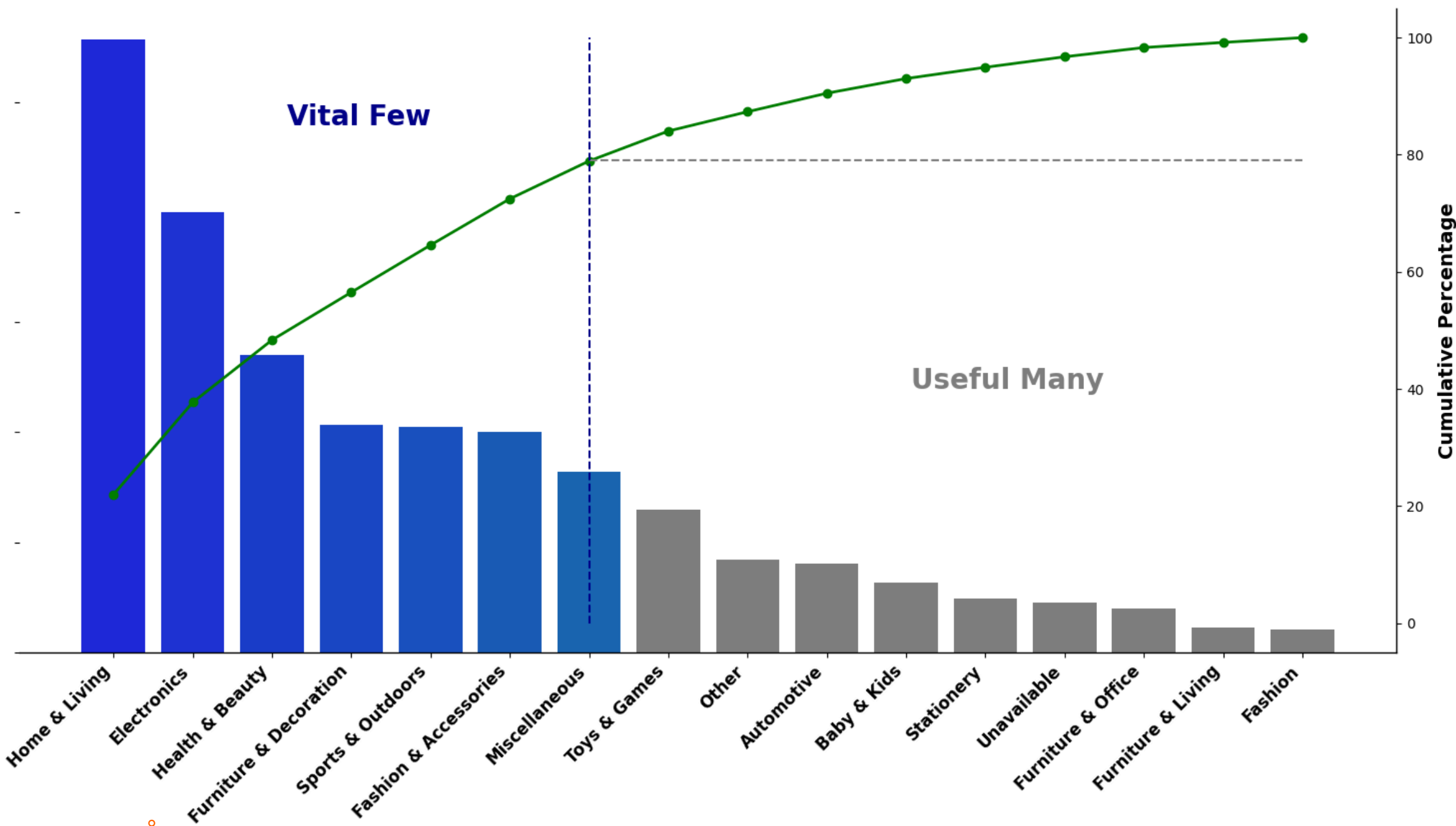
Customer State	Acre	0	1	0	1	0	1	1	0	1	10	1	0	0	0	0	0	2	1	0	4	0	61
	Alagoas	1	9	2	4	0	9	4	0	0	36	39	1	0	9	0	1	10	13	0	11	0	283
	Amapá	0	2	0	0	0	1	1	0	0	3	2	0	0	1	0	0	3	1	0	4	0	55
	Amazonas	0	1	1	0	0	2	1	0	0	14	14	0	0	0	0	0	5	9	0	4	0	104
	Bahia	0	70	3	36	23	8	10	4	3	397	160	2	0	27	0	7	42	142	1	87	0	2619
	Ceará	0	14	9	16	6	12	9	2	0	80	72	4	0	17	1	3	15	55	0	23	0	1072
	Distrito Federal	0	18	1	51	1	29	27	7	4	217	168	1	0	16	0	1	50	107	0	76	0	1555
	Espírito Santo	0	11	3	19	8	5	13	2	0	204	116	1	0	17	0	0	26	107	1	55	1	1618
	Goiás	0	8	1	31	3	34	16	10	1	160	91	2	0	12	0	0	23	119	0	59	1	1637
	Maranhão	1	16	2	8	2	23	17	3	0	73	47	1	0	5	1	0	15	32	0	25	0	537
	Mato Grosso	0	5	2	15	6	12	4	7	2	58	77	1	0	1	0	0	25	33	0	28	0	760
	Mato Grosso do Sul	0	2	1	7	0	8	3	5	0	70	58	0	0	2	0	0	16	35	0	31	0	532
	Minas Gerais	1	53	5	106	47	91	32	21	3	1673	954	2	1	71	2	1	196	593	0	461	0	8297
	Paraná	0	11	5	28	15	16	9	8	5	385	762	0	1	9	2	4	152	217	1	300	1	3428
	Paraíba	0	10	1	8	0	7	4	2	0	49	43	1	0	8	1	3	8	25	0	20	0	379
	Pará	0	8	3	11	3	11	5	3	0	88	57	1	0	5	0	1	21	39	0	25	1	770
	Pernambuco	0	33	7	10	6	11	25	6	0	114	93	2	0	23	1	0	17	95	0	41	4	1255
	Piauí	0	9	3	8	2	2	5	0	0	32	27	0	0	8	1	0	12	26	0	19	0	362
	Rio Grande do Norte	0	10	1	6	1	1	3	1	0	32	19	1	0	4	0	24	8	23	0	14	0	373
	Rio Grande do Sul	0	22	2	22	20	14	17	6	1	301	659	0	0	13	0	4	305	218	1	324	0	4033
	Rio de Janeiro	0	86	10	101	61	61	41	13	9	1284	1100	10	0	46	1	2	268	1036	2	508	1	9285
	Rondônia	0	3	1	0	1	1	0	3	0	15	18	0	0	3	0	0	10	13	0	9	0	199
	Roraima	0	1	0	0	0	0	0	0	0	2	5	0	0	0	0	0	0	0	0	1	0	35
	Santa Catarina	0	16	2	22	13	10	19	5	1	278	432	0	0	9	1	1	129	123	0	287	0	2557
Sergipe	0	9	3	3	1	0	6	0	0	36	24	1	0	7	0	0	8	21	1	20	0	219	
São Paulo	0	147	25	317	109	141	128	35	19	2873	3303	9	6	105	2	5	722	1449	8	1447	2	34601	
Tocantins	0	2	0	5	2	5	2	0	0	17	17	0	0	1	0	0	4	15	0	12	0	227	
		Amazonas	Bahia	Ceará	Distrito Federal	Espírito Santo	Goiás	Maranhão	Mato Grosso	Mato Grosso do Sul	Minas Gerais	Paraná	Paraíba	Pará	Pernambuco	Piauí	Rio Grande do Norte	Rio Grande do Sul	Rio de Janeiro	Rondônia	Santa Catarina	Sergipe	São Paulo
		Seller State																					



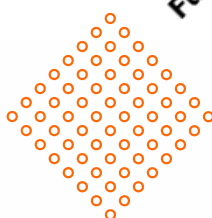
Business Recommendations

80% of Revenue Comes from **7 Core Product Categories**

Pareto Chart: Top 15 Product Categories by Quantity and Cumulative Percentage



Based on this Pareto visualization, 80% of the revenue comes from 7 core product categories: **Home & Living, Electronics, Health & Beauty, Furniture & Decoration, Sports & Outdoors, Fashion & Accessories, and Miscellaneous**. This highlights the importance of focusing on the "vital few" categories that contribute the most to revenue, while other categories contribute significantly less.





Business Recommendations

1. Expand Focus to Key States

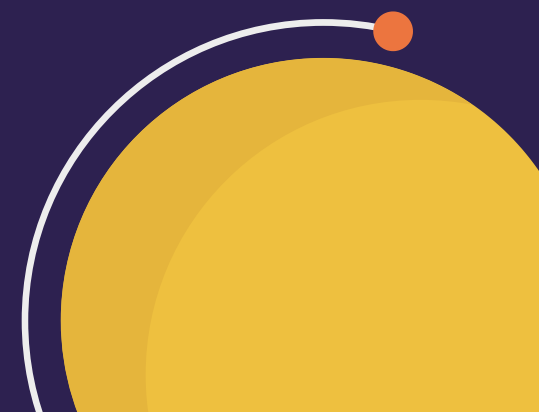
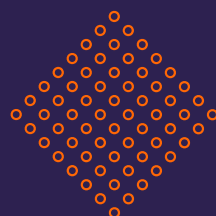
Target increased sales in alternative states such as Minas Gerais, Paraná, Rio Grande do Sul, Rio de Janeiro, and Santa Catarina, in addition to São Paulo. This will help reduce order fulfillment failures and optimize distribution networks in these regions, ensuring broader market reach and more efficient product delivery.

2. Prioritize High-Demand Product Categories

Based on the Pareto analysis, focus on high-demand product categories such as Home & Living, Electronics, Health & Beauty, Furniture & Decoration, Sports & Outdoors, Fashion & Accessories, and Miscellaneous. Concentrating on these categories in the targeted states will maximize revenue and reduce stockouts that may disrupt order fulfillment.

3. Conduct In-Depth Customer Satisfaction and NPS Analysis

Perform a detailed analysis of customer satisfaction and Net Promoter Score (NPS) to understand customer experiences in the e-commerce platform. By identifying areas for improvement in user experience, delivery, and product quality, you can enhance customer loyalty and increase repeat purchases.



Dashboard Preview

